



LOGISTICS PRODUCT

Keywords: Container Orchestrator, Cost Saving

PODTetris: Rule-Based Pod Rescheduling to Reduce Node Count in Kubernetes Clusters

Abstract

In Kubernetes clusters, workloads can end up spread across nodes in a suboptimal way, especially when only a few applications scale automatically. This often leads to running **more nodes than necessary**, increasing cost and wasting capacity.

This experimental thesis proposes **PODTetris**, a solution that periodically “rearranges” pods inside the cluster. The goal is to **pack workloads onto fewer nodes** by moving pods in a controlled manner, while still respecting existing placement rules (for example, constraints that keep certain pods together or apart). The approach is based on **simple, configurable, and easy-to-implement rules**, designed to be practical in real engineering environments.

The work includes designing the rules, building a prototype, and validating it through test and real production scenarios, measuring node reduction and the impact on application stability.

Work steps

1. Problem framing and constraints

- Understand why inefficient spreading happens and which placement rules must never be violated.

2. Define goals and success metrics

- Examples: node count reduction, number of pod moves, stability during the process.

3. Design the rescheduling rules

- Decide *when* to act, *which* pods are eligible, and *how* to choose the new placement.
- Make rules **configurable** without code changes.

4. Build the prototype

- Implement a component that observes cluster state, computes a rearrangement plan, and applies moves safely.

5. Test and experimental validation

- Create test scenarios (few workloads, different placement constraints).
- Compare before/after and analyze results and limitations on real production environments.

6. Conclusions and future improvements

- Identify where it works best and what could be extended next.